

For questions 1-2, consider the expression $15y^3 - 10y^2 - 25y$.

1. Write an equivalent expression that uses the greatest common monomial factor.

$$5y(3y^2 - 2y - 5)$$

2. Write an equivalent expression that does not use the greatest common monomial factor.

Answers may vary. Example: $5(3y^3 - 2y^2 - 5y)$

For questions 3-6, write an equivalent expression for the given expression that uses the greatest common monomial factor.

3. $3f^3g^2 + 15fg^3 - 27f^2g^2$

$$3fg^2(f^2 + 5g - 9f)$$

4. $63s^5h^5 + 42s^3h^6 + 35s^4h^2$

$$7s^3h^2(9s^2h^3 + 6h^4 + 5s)$$

5. $15d^3b^3 - 60db^2$

$$15db^2(d^2b - 4)$$

6. $12x^7y^6 - 108x^3y^5 - 84x^6y^4$

$$12x^3y^4(x^4y^2 - 9y - 7x^3)$$

For questions 7-10, find the greatest common monomial factor of the expression.

7. $13a^3 - 39a^2b$

$13a^2$

8. $12q^2c^2 + 30qc + 42q^8c^4$

$6qc$

9. $10r^7m^5 - 55r^2m^3 + 40r^5m$

$5r^2m$

10. $18x^5y^3 - 40x^6y^6 - 38xy$

$2xy$